Mechanics and Gravitation

Celestial Mechanics and the Architecture of the Cosmos

Sarthak Arora

UG 3rd Year

Astrae | IISc

Gravity Across Scales

Although gravity is the weakest fundamental interaction, it exclusively governs the structure and evolution of the Universe. From artificial satellites and planetary systems to stellar clusters and spiral galaxies, the same inverse-square law explains both local orbits and the vast, large-scale structures of the cosmos.

Newtonian Gravity & Conservation

- **Inverse-Square Law:** Gravity is a central, conservative force always directed along the line joining two interacting masses, responsible for stable planetary orbits.
- **Conservation of Energy:** The total mechanical energy (Kinetic + Potential) remains strictly constant in any isolated gravitational system.
- **Conservation of Linear Momentum:** In the absence of external forces, the centre of mass of a system moves uniformly.
- **Conservation of Angular Momentum:** Angular momentum is conserved for all central forces, dictating the geometric stability of orbital planes.

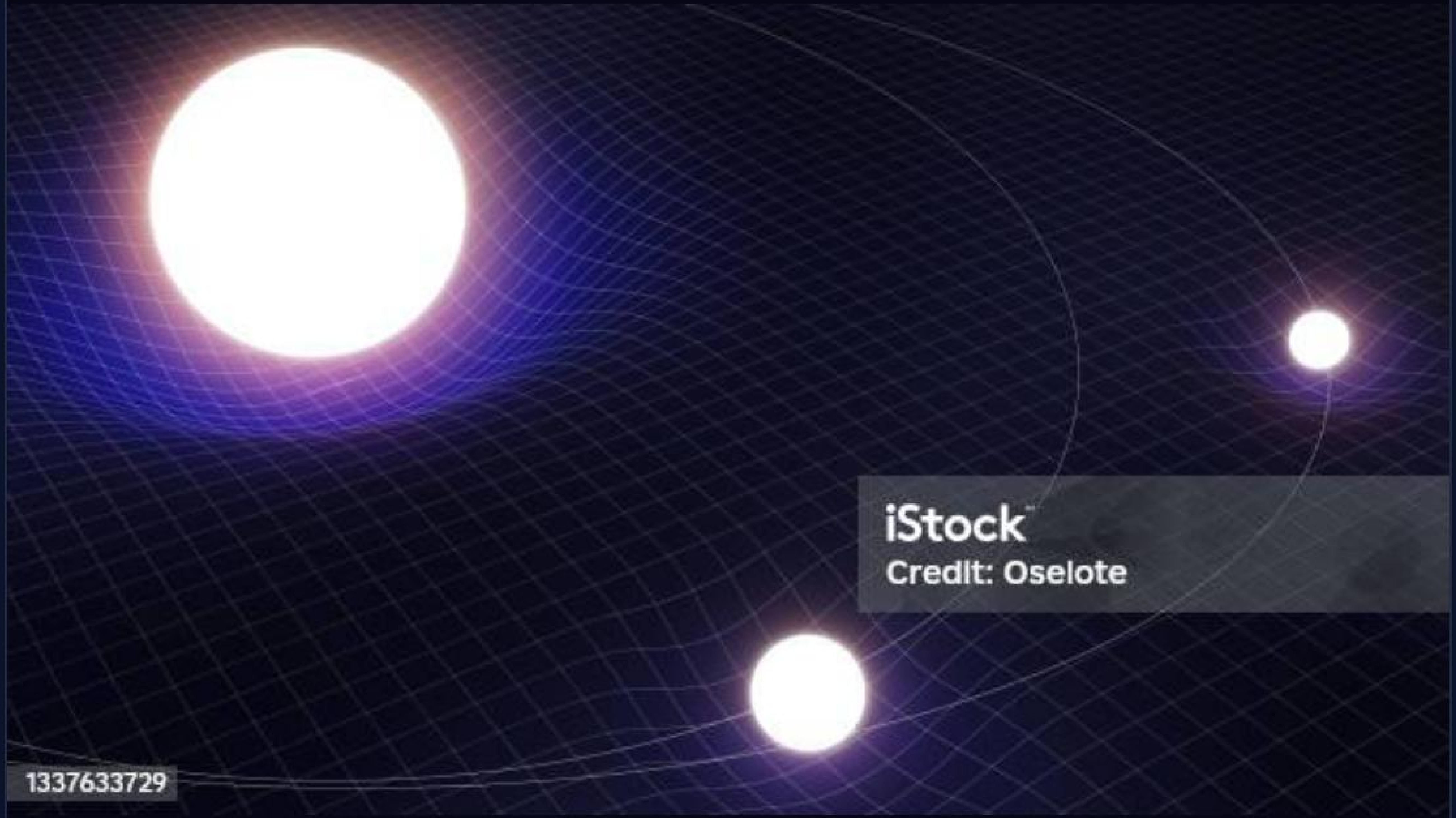
The Two-Body Problem

Instead of solving complex coupled equations for two massive bodies interacting gravitationally, we reduce the problem to the motion of a single particle.

By introducing a **reduced mass** moving in a gravitational effective potential, we mathematically simplify celestial dynamics.

Why is this important?

The Earth-Moon system, planetary orbits around the Sun, and even complex binary star systems can all be analytically treated using exactly the same unified mathematical framework.



Conic Sections as Orbital Solutions

The value of orbital eccentricity uniquely determines the fundamental type and bound state of a trajectory.

ORBIT TYPE	ECCENTRICITY (E)	TOTAL ENERGY (E)	MOTION TYPE
Circle	$e = 0$	Negative	Bound / Periodic
Ellipse	$0 < e < 1$	Negative	Bound / Periodic
Parabola	$e = 1$	Zero	Unbound / Escape
Hyperbola	$e > 1$	Positive	Unbound / Escape

Kepler's Laws

First Law

Law of Ellipses: Planetary orbits are purely elliptical with the Sun occupying one of the principal focal points, breaking the ancient paradigm of perfect circles.

Second Law

Law of Equal Areas: The radius vector sweeps out equal areas in equal times, an observation that directly emerges from the conservation of angular momentum.

Third Law

Law of Harmonies: The square of the orbital period is strictly proportional to the cube of the semi-major axis, allowing mass determination of distant stars.

Orbital Mechanics Equations

Vis-Viva Equation

Orbital speed varies throughout an elliptical orbit. Combining energy and angular momentum principles allows velocity computation at any orbital radius.

$$v^2 = GM\left(\frac{2}{r} - \frac{1}{a}\right)$$

Escape Velocity

The minimum velocity required to completely escape a massive body's gravitational potential field, independent of the escaping object's mass.

$$v_e = \sqrt{\frac{2GM}{R}}$$

Gravitational Wells: Escape Velocities

> *The deeper the gravitational potential well, the larger the escape velocity required. This fundamental property dictates atmospheric retention, planet formation capabilities, and highly influences spacecraft mission design.*

Origin: The Nebular Hypothesis

1. Cloud Collapse

Gravitational collapse of a giant, rotating interstellar molecular cloud triggers the formation sequence.

2. Disk Flattening

Conservation of angular momentum causes the collapsing cloud to flatten into a rapidly spinning protoplanetary disk.

3. Proto-Sun Ignition

Extreme pressure and temperature at the core ignite nuclear fusion, birthing the Proto-Sun at the center.

4. Planetesimal Accretion

Dust coagulation and violent collisions form planetesimals, eventually merging into mature planetary bodies.

Solar System Architecture

Terrestrial Planets

Formed inside the frost line. Rocky composition, high density, and localized atmospheric envelopes (e.g., Earth, Mars).

Giant Planets

Formed beyond the frost line where volatiles freeze. Massive gaseous/ice envelopes and strong magnetic fields.

Small-Body Reservoirs

Asteroid Belt, Kuiper Belt, and Oort Cloud. Primitive cosmic fossils preserving planetary formation history.

Resonances & Migration

The Solar System is not a static arrangement but a dynamically evolving N-body system. Mutual perturbations constantly shape celestial architecture.

- **Orbital Resonances:** Occur when orbital periods form simple integer ratios (e.g., Pluto/Neptune 3:2), providing long-term stability or triggering chaotic migrations.
- **The Nice Model:** Postulates that giant planets originally occupied highly compact inner orbits before gravitational instability scattered them outward.

Understanding cosmic evolution mandates numerical N-body simulations.

Questions?

Thank you for your attention.

Image Sources

https://as1.ftcdn.net/v2/jpg/01/18/87/40/1000_F_118874087_414nX1auMugFwRwK79je2oO26rftihGd.jpg

Source: www.ukposters.co.uk

https://media.istockphoto.com/id/1337633729/photo/gravity-simulation-in-space-space-and-time-in-the-universe-mass-affects-space-and-time-theory.jpg?s=612x612&w=is&k=20&c=wL2TUfKc2N6AHMh28wCfeuzHSyll6jiSmUUJAVnzYNM=

Source: www.istockphoto.com

https://media.istockphoto.com/id/2057625445/photo/earth-and-mars.jpg?s=612x612&w=0&k=20&c=lNQ6zZYf2gJPV3s_3ts51EG633BVrxHgnvF0xEN3PhU=

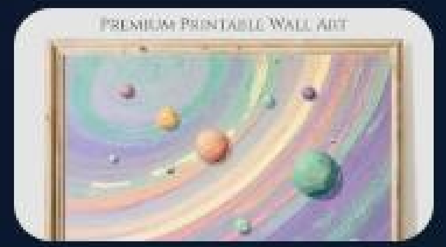
Source: www.istockphoto.com

https://science.nasa.gov/wp-content/uploads/2024/03/jupiter-marble-pia22946-16x9-1.jpg?w=1920

Source: science.nasa.gov

https://static.vecteezy.com/system/resources/previews/067/802/387/non_2x/asteroid-field-in-space-a-high-resolution-render-of-gray-rocks-floating-in-the-black-void-ideal-for-science-fiction-backgrounds-transparent-background-png.png

Source: www.vecteezy.com



https://i.etsystatic.com/6707196/r/il/8540d3/7310575979/il_570xN.7310575979_ocri.jpg

Source: www.etsy.com